

# Oliver Neal

Gameplay Programmer · Unity (C#)

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Middlesbrough, UK · Right to work in the UK · Open to relocation and remote work

## PROFILE

Gameplay programmer specialising in Unity (C#) systems work: state-machine-driven characters and enemy AI, local co-op and multiplayer plumbing, and the profiling that keeps a scene inside its frame budget.

First Class Honours in BSc (Hons) Games Development from Teesside University, winner of Best Games Programming and Development Artefact at ExpoTees 2026, and currently a gameplay programmer at Shiverbug Studios on *Out of Water*. Open to gameplay, systems or AI programming roles from September 2026.

Making games in Unity since 2018, building them consistently since 2023. Nine shipped projects, most playable in the browser, with source on GitHub and technical breakdowns at [oliverneal.dev](https://oliverneal.dev).

## EXPERIENCE

**Gameplay Programmer (Internship)** · Shiverbug Studios

Jul to Sep 2026

- Own the character and enemy gameplay code on *Out of Water*, an original-IP 3D splitscreen platformer collectathon for two players, working in engine daily.
- Rebuilt both playable characters and the enemies from single-file controllers onto component-driven finite and hierarchical state machines, and built the co-op moveset on top.
- Ran the performance pass: GPU-driven rendering, quality tiers, LOD and culling, measured branch to branch with a profiling tool I wrote for the team.
- Built the splitscreen plumbing (per-player device routing, cameras, character swapping) and a UI Toolkit front end on a shared token system: HUD, character select, pause, save slots and a controls tutorial.
- Handled presentation and tooling: water, caustics and bubble shaders in Shader Graph and hand-written URP HLSL, a decoupled FMOD layer the audio team plugs into directly, and the blockout interactables and editor tooling the level designers build puzzles from.
- Designed and shipped the studio's public website, [shiverbugstudios.com](https://shiverbugstudios.com), alongside gameplay work.

Unity 6 · C# · FSM/HSM · Performance · UI Toolkit · FMOD · Cinemachine · Shader Graph · HLSL

## TECHNICAL SKILLS

**Languages & engines:** C#, Unity 6 (2D/3D), Shader Graph, HLSL, Unreal Engine 5, Python

**Gameplay & AI:** gameplay programming, systems design, finite & hierarchical state machines, AI navigation (NavMesh), local co-op & splitscreen, Unity Input System

**Performance:** profiling and frame analysis, time-sliced & multi-rate simulation, GPU-driven rendering, LOD and culling, quality tiers, runtime NavMesh baking

**Audio, UI & presentation:** FMOD, UI Toolkit, Unity UI, Cinemachine, particle VFX

**Multiplayer & services:** Netcode for GameObjects, Unity Authentication, Relay, Cloud Save

**AI-assisted development:** Claude Code and Codex as accelerators in a spec-first workflow, with architecture, implementation direction and code review staying mine

**Tools & workflow:** Git and Git LFS, branch workflow on a shared team repo, Visual Studio, JetBrains Rider, Figma, Trello, technical documentation

## HIGHLIGHTED SOLO PROJECTS

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**Cyber Station** · Unity 3D · C# · AI navigation · simulation Apr 2026

- Neon station management game, 86 scripts and ~17k lines solo: hundreds of passengers run off one fixed 10 Hz tick, picking facilities by lowest estimated total delay (queue wait plus real NavMesh walk time, both in seconds), with six live ratings feeding back into the spawn rate.
- Final-year artefact: scored 90/100 and won Best Games Programming and Development Artefact at ExpoTees 2026.  
Breakdown: [oliverneal.dev/cyber-station.html](https://oliverneal.dev/cyber-station.html) · Play: [olivernealdev.itch.io/cyber-station](https://olivernealdev.itch.io/cyber-station) · Source: [github.com/OliverNealDev/CyberStation](https://github.com/OliverNealDev/CyberStation)

**Tile Turfer** · Unity 2D · C# · procedural generation · enemy AI Jan 2026

- Turf-war shooter where one territory percentage drives difficulty, fire rate and enemy AI at once, across procedurally generated caves with runtime 2D NavMesh baking.  
Breakdown: [oliverneal.dev/tile-turfer.html](https://oliverneal.dev/tile-turfer.html) · Play: [olivernealdev.itch.io/tile-turfer](https://olivernealdev.itch.io/tile-turfer) · Source: [github.com/OliverNealDev/TileTurfur](https://github.com/OliverNealDev/TileTurfur)

**Minimalists** · Unity 3D · C# · enemy AI · RTS Aug 2025

- Node-conquest RTS whose enemy AI gets no cheats and no extra information: it commits through the same percentage interface the player uses, sizing each attack from what the target actually holds, across several distinct AI personalities.  
Breakdown: [oliverneal.dev/minimalists.html](https://oliverneal.dev/minimalists.html) · Play: [olivernealdev.itch.io/minimalists](https://olivernealdev.itch.io/minimalists) · Source: [github.com/OliverNealDev/Minimalists](https://github.com/OliverNealDev/Minimalists)

**Wanted** · Unity 6 · C# · stealth AI · one-week game jam May 2025

- Top-down stealth survival game built in one week for the Tyne to Game jam: a single detection meter drives three-state officer AI, escalating patrols and helicopter passes on a runtime-baked 2D NavMesh.  
Breakdown: [oliverneal.dev/wanted.html](https://oliverneal.dev/wanted.html) · Play: [olivernealdev.itch.io/wanted](https://olivernealdev.itch.io/wanted) · Source: [github.com/OliverNealDev/Wanted](https://github.com/OliverNealDev/Wanted)

**Unicellular** · Unity · C# · FSM · optimisation Feb 2025

- Idle simulation stress-tested at 1,000 and 10,000 concurrent entities, kept smooth by a decoupled multi-rate tick and time-sliced proximity AI.  
Breakdown: [oliverneal.dev/unicellular.html](https://oliverneal.dev/unicellular.html) · Play: [olivernealdev.itch.io/unicellular](https://olivernealdev.itch.io/unicellular)

**3D Netcode Tank Game** · Unity 3D · C# · netcode · multiplayer Jan 2025

- Server-authoritative multiplayer tank shooter: Relay-hosted sessions, cloud authentication, a decoupled turret aiming rig, and a 64-ray hitscan resolved on the server.  
Breakdown: [oliverneal.dev/tank-game.html](https://oliverneal.dev/tank-game.html) · Source: [github.com/OliverNealDev/NetcodeTankGame](https://github.com/OliverNealDev/NetcodeTankGame)

**Pinnable** · Unity 2D · C# · arcade · physics Mar 2024

- Arcade score-attack catch game with a self-escalating spawn director: hazards unlock on a survival timer, then scale their own spawn weight with score, so difficulty tracks skill rather than a fixed curve.  
Breakdown: [oliverneal.dev/pinnable.html](https://oliverneal.dev/pinnable.html) · Play: [olivernealdev.itch.io/pinnable](https://olivernealdev.itch.io/pinnable) · Source: [github.com/OliverNealDev/Pinnable](https://github.com/OliverNealDev/Pinnable)

## TEAM PROJECTS

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**Space Bar Simulator** · Unreal Engine 5 · Blueprints · team of 10 Jan 2026

- Retro-futurist first-person bar simulator for Beta Arcade, Teesside's studio-simulation module: owned the player controller on Enhanced Input, a component-based interaction system behind a Blueprint interface, camera-blended stations and the UI computer terminal.  
Breakdown: [oliverneal.dev/space-bar.html](https://oliverneal.dev/space-bar.html)

**Candy Corporate** · Unity 2D · small team · pixel art Apr 2024

- Top-down extraction game inspired by Lethal Company; led level design, tile creation and pixel-art integration.  
Play: [jurgenwithwings.itch.io/candycorporate](https://jurgenwithwings.itch.io/candycorporate)

## EDUCATION

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**BSc (Hons) Games Development** · Teesside University Jul 2026

- First Class Honours, 80% overall; final-year project scored 84%, its artefact 90/100.

**FdSc Games Technologies** · Newcastle College University Centre Jul 2025

- 2:1 (Year 1 Distinction, Year 2 Merit).

**BTEC Level 3 National Diploma, Creative Digital Media** · Newcastle College 2023

- Distinction.